

Chapter 165

Resolve FX Stylize

The plugins found in this category all enable different ways of creating artistic modifications to the image.

Contents

Abstraction (Studio Version Only)	3577	Rosette Controls.....	3586
Main Controls.....	3577	Kaleidoscope Controls.....	3586
Quantization Controls.....	3577	Pencil Sketch in DaVinci Resolve (Studio Version Only).....	3587
Draw Edge Controls.....	3577	Prism Blur	3588
Blanking Fill	3577	Scanlines	3588
Source.....	3578	Appearance.....	3588
Fill Extent.....	3579	Color.....	3588
Fill Appearance.....	3580	Composite.....	3589
Drop Shadow.....	3580	Sky Replacement	3589
Defocus Background	3580	Selecting the Sky to be Replaced.....	3590
Adjust Mask.....	3581	Selecting the Replacement Sky.....	3590
Effects.....	3582	Sky Mask Adjustments.....	3591
Advanced Options.....	3582	Source Sky Appearance.....	3591
Drop Shadow	3582	Artificial Sky.....	3591
Edge Detect	3583	Sky Position.....	3592
Main.....	3583	Sky Integration.....	3593
Detection.....	3583	Foreground Appearance.....	3593
Filters.....	3584	Sky Replacement Workflow.....	3593
Advanced Options.....	3584	Stylize (Studio Version Only)	3596
Emboss	3585	Tilt-Shift Blur (Studio Version Only)	3597
Channels.....	3585	Main Controls.....	3597
Mirrors	3586	Lens Iris.....	3597
Main Controls.....	3586	Depth of Field.....	3597
Individual Controls.....	3586	Watercolor (Studio Version Only)	3598

Abstraction (Studio Version Only)

A deceptively powerful filter that lets you create a wide range of cartoon-like renders by simplifying an image into adjustable pools of similar color with optional outlines.

Main Controls

These controls create the foundation of this effect.

- **Pre Blur:** Simplifies the image by blurring unwanted details prior to this filter taking effect.
- **Abstraction Strength and Iterate Abstraction:** These parameters work together to then smooth the image, creating pools of simplified color within the details of the image.

Abstraction Strength: Must be higher than 0 for the Iteration No. parameter to have an effect. Higher Abstraction Strength simplifies image detail even more by averaging adjacent regions of color together.

Iterate Abstraction: Reduces the amount of small detail in the image even further by “widening” adjacent areas of similar color to blend together as one.

Quantization Controls

These controls simplify the pools of color even more aggressively by quantizing the bit depth used to stylize the image. The result is an even sharper flattening of each region of color.

- **Quantization:** A checkbox turns this function on.
- **Steps:** Raising this parameter lets you subdivide each region of color into a greater number of separate regions.
- **Softness:** Lets you blur the border between each level of color.

Draw Edge Controls

Adds an edge to the regions of color created by this effect.

- **Draw Edge:** Enables edge drawing.
- **Edge Strength:** Raising this value results in thicker and better-defined edges.
- **Edge Detection Threshold:** Raising this value restricts edges to only the boldest edge details in the picture.

Blanking Fill

This plugin is specifically designed to quickly fill black frame blanking with a stylized image derived from the clip itself, to make blanking less intrusive for viewers in documentaries and news segments. In the following example, the Blanking Fill plugin is used to add image to the left and right of “pillarboxed” standard definition 4:3 video that appears within a High Definition or Ultra High Definition 16:9 aspect ratio.



(Top) Original standard definition image (Bottom) Image with default Blanking Fill applied

A variety of controls let you easily customize this effect to suit your own purposes.

Source

These controls let you transform the clip in different ways, which also affects how that clip is used to fill the blanking area.

- **Zoom:** Zooms the clip while keeping it within the native domain of definition. Good for quickly cropping unwanted pixels around the edges so they're hidden and aren't used for blanking fill.
- **Crop Left and Right:** Crops both the left and right edges of the image and increases the area at the sides with blanking fill.
- **Same Left/Right Sliders:** On by default, locks both sliders together so moving one moves the other by a mirrored amount.
- **Crop Top and Bottom:** Crops both the top and bottom edges of the image and increases the vertical area with blanking fill.
- **Same Top/Bottom Sliders:** On by default, locks both sliders together so moving one moves the other by a mirrored amount.
- **Center:** If checked, re-centers the image area if a sizing operation has offset it.

Fill Extent

These controls let you choose how the current duplicate of the image is stretched to fill the blanking area.

- **Zoom Mode:** There are three options:

Stretch to Timeline: Automatically warps the image to stretch it to fit the full frame. Keeps features in the blanking fill area level with where they appear in the original image.

Zoom to Timeline: Automatically zooms into the image to fit the full frame. Results in the clip looking “inset” from the blanking fill image.

Manual: Reveals Expand and Aspect sliders that let you manually choose how much to zoom and stretch the image to fit the blanking fill area.

- **Warp Top Layer:** lets you use onscreen controls to choose a section of the edge of an image to stretch out to fill blanking in the frame. In this mode, there are two sets of onscreen controls you can use to customize the result:

A set of outer handles let you choose how far out to warp the edges of the image to fill whatever blanking there is. These default to the project frame size.

A set of inner handles lets you choose how much of the image you want to stretch out. These default to the current title safe boundary. If you customize these, choosing too narrow an edge results in a more extreme warping and stretching effect, while choosing a wider edge to stretch looks more natural, but affects more of the frame.

- **Expand:** (Only appears when Zoom Mode is set to Manual) Zooms the image.
- **Aspect:** (Only appears when Zoom Mode is set to Manual) Stretches the image.



The original 4:3 image edited into a 16:9 timeline with black pillar boxing to the left and right as a result



The image with Blanking Fill set to Warp Top Layer, stretching the left and right edges to fill the blanking area

Fill Appearance

These controls let you change the look of the image being used to fill the blanking area.

- **Blend Edges:** Lets you feather the edges where the original image meets the blanking fill area.
- **Blur Background:** Lets you choose how much or how little to blur the image that fills the blanking fill area.
- **Fade Amount:** Lets you fade or tint the blanking fill area. At 0, there's no fade applied. At 1.000, the blanking area is filled with a solid color defined by the Fade Color control.
- **Fade Color:** A color picker/eyedropper combination that lets you choose a color with which to fade, tint, or fill the blanking fill area.

Drop Shadow

These controls let you add a drop shadow to “lift” the foreground image against the blanking fill image.

- **Shadow Strength:** Lets you choose how solid the drop shadow appears.
- **Drop Angle:** Lets you choose the angle the drop shadow appears at. At 0 the drop shadow is centered.
- **Drop Distance:** Offsets the drop shadow farther in the direction of the Drop Angle
- **Blur:** Softens the drop shadow.
- **Color:** Tints the drop shadow.

Defocus Background

The Defocus Background effect lets you blur only the background of an image and provides a detailed set of tools to mimic shooting with a shallow depth of field.

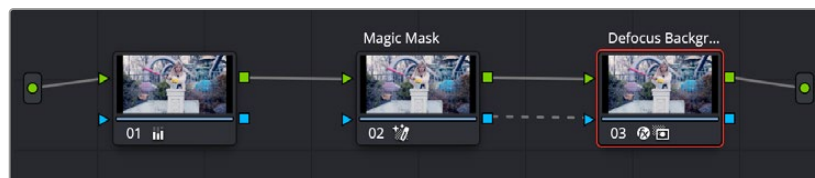


The original image with a busy, distracting background



Using the Defocus Background effect to blur the background, making it less distracting and mimicking a shallower depth of field

The Defocus Background effect does not qualify or isolate the foreground subject. In order to use the Defocus Background, you must first isolate the foreground subject using an effect or mask using another tool like Magic Mask or a Power Window. You can then apply Defocus Background either to the same node or connect the key output from a previous node.



Connecting the key output of the Magic Mask to the Defocus Background effect; Defocus Background requires a key generated from another tool like Magic Mask or a Power Window to work.

Adjust Mask

These controls let you adjust the mask that separates the sharply focused areas from the areas you want to blur.

- **Mask Preview:** Checking this box will bring up a black and white preview of your mask. The white areas of the mask will stay sharp, while black areas of the mask are what will be blurred using the mask controls below. This is convenient to have up while you're perfecting the mask, then disabled to see the output of the Effects and Advanced options.
- **Mask Invert:** Checking this box will invert the mask, making the white areas black and vice-versa.
- **Mask Blur:** This control adjusts the size of the Blur around the mask, letting you smoothly blend the foreground and background together.
- **Mask Contrast:** This controls how strong the divide is between the focused and blurred regions of the matte.
- **Adjust Pivot:** This allows you to set where the divide is between the focused and blurred regions of the matte.



The Mask Preview option shows where the defocus operations will take place (black areas).

Effects

These controls set the main characteristics of the blurred area.

- **Blur:** This slider controls the size and how much blur to apply to the blurred area.
- **Saturation:** This slider controls the saturation of the blurred area. 1.000 (default) is full saturation. A value of 0.000 makes the blurred areas monochrome.
- **Colorize:** This slider controls the amount of color to tint the blurred areas. 0.000 is the default (no colorization).
- **Color:** This is the color picker for the Colorize operation above.

Advanced Options

These controls let you adjust some of the physical characteristics of the blurred area.

- **Blur Type:** This lets you choose the blurring algorithm used to make the blur. Either Lens blur or Gaussian blur are options.
- **Anamorphism (Lens Blur):** This slider lets you set the aspect ratio of the blur to match different lens types.
- **Highlights (Lens Blur):** Lets you adjust how the highlights of the image affect the blur, dilating or eroding the image.

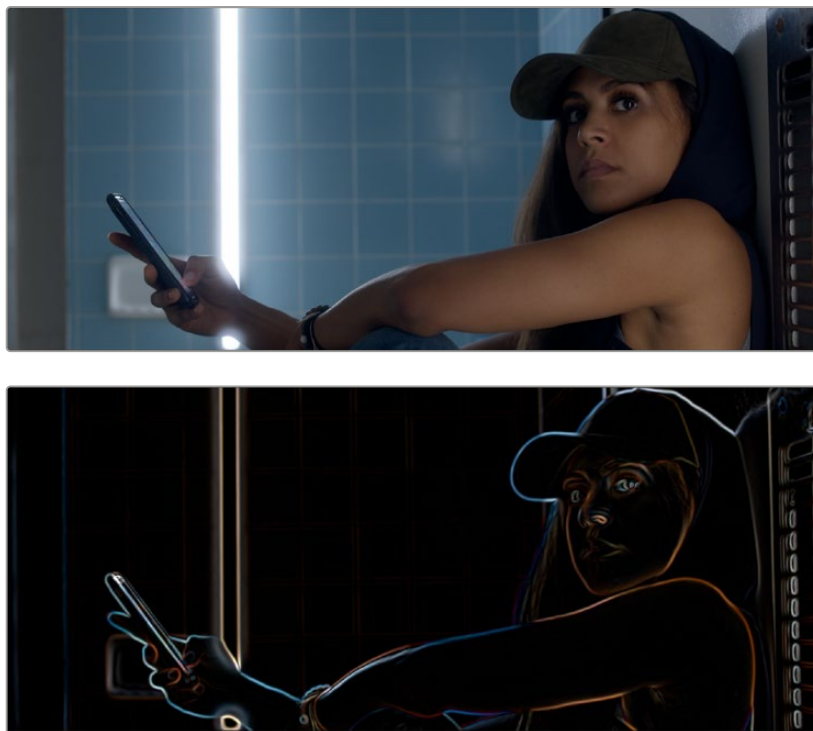
Drop Shadow

Lets you add a simple drop shadow to any clip, using that clip's native or generated alpha channel to create the shape of the shadow.

- **Shadow Strength:** Lets you choose how solid the drop shadow appears.
- **Drop Angle:** Lets you choose the angle the drop shadow appears at. At 0 the drop shadow is centered.
- **Drop Distance:** Offsets the drop shadow farther in the direction of the Drop Angle
- **Blur:** Softens the drop shadow.
- **Color:** Tints the drop shadow.

Edge Detect

An edge detection effect that isolates high-contrast boundaries with options for customizing which edges create outlines. With the edges isolated, this allows you to sharpen soft clips or to create a glow around a subject.



(Top) Original image, (Bottom) Image with default edge detect applied

Main

- **Mode:** Lets you choose between RGB and Grayscale edges. RGB is the default. RGB finds edges based on color and brightness, while Grayscale detects edges based solely on brightness changes.
- **Edge color:** (only enabled if Mode is set to Grayscale) Lets you colorize the edge outlines that are generated.

Detection

These controls determine the strength and parameters of the detected edges.

- **Edge Width:** Lets you set the thickness of the resulting edges.
- **Brightness:** Lets you set how bright you want the edges to be.
- **Gamma:** Applies a gamma curve to the edge detection. A gamma below one adds brightness to mid-strength edges, making more edges “detected,” while a gamma above one darkens mid-strength edges, reducing the edges to all but the strongest detected.
- **Blur:** Softens the final result, allowing the edges to blend in more naturally.
- **Edge Mask Overlay:** This checkbox overlays the edges on top of the current frame to observe the filter’s results. Unchecking this box shows only the detected edges.

Filters

These controls determine how the edges are blended with the original frame.

- **Preprocess:** Lets you choose which method of de-noising filtering to use. The default is None.
 - Filter:** Reduces texture and filters out detail at the chosen Scale parameter below.
 - Flatten:** Flattens smooth areas of the image, leaving edges only where they divide between other flat regions in the image. The Strength parameter below controls the amount of this smoothing.
- **Half-Edges Only:** Edges are detected between regions of different brightness. This option lets you determine which side of the edge to keep. The default is None (both sides of the edge are kept).
 - Keep Light Side:** Edge selects only the brighter half of the edge. Useful, for example, in creating a light-wrap effect on a person's face as only the lighter (face side) of the edge is selected, limiting the increased brightness of the light wrap to the face only and not the background. The Half-Edge Gain slider allows you to further boost the brightness of the edge, separately from the main Brightness control.
 - Keep Dark Side:** Edge selects only the darker half of the edge. Useful, for example, in creating a glow effect on a person's face as only the darker (background side) of the edge is selected, allowing a halo to form around the face, but not on it. The Half-Edge Gain slider allows you to further boost the brightness of the edge, separately from the main Brightness control.

Advanced Options

These controls allow you to select some advanced parameters for the effect.

- **Normalize Per Frame:** Performs a brightness adjustment on each incoming frame to a consistent level, causing edges to remain visible as the scene changes brightness or flickers. Use this setting if your scene has content that has constant changes in brightness levels, otherwise uncheck this box.
- **Clamp Brightness:** Keeps the output edges from becoming too bright if the contrast of the content is too high.

For example, if you wanted to make a slight glow around the woman in the frame below using edge detect:

- Select RGB Edges, Check the Edge Mask Overlay box, and adjust the Blur to about .5 to blend them into the frame subtly.
- Next select Flatten from the Preprocess control to smooth out the fine details in the image, ensuring that only the highlights from the woman's hat, arm, and phone are detected, as well as the bright fluorescent light on the wall. These are what we want to "glow." Adjust the Strength slider to taste.
- Then select "Keep Dark Side" in the Half-Edges Only box. This ensures that the glow radiates out from the edges and not onto the woman and the light. Adjust the Half-Edge Gain slider to taste.



The shot before the Edge effect



The Edge Detect effect only on the same image with the parameters described above



The Edge Detect result; notice the glow coming from the light and around the woman's hat, arm, and phone.

Emboss

An emboss effect with several options to create different kinds of outline and emboss effects.

- **Emboss Style:** This pop-up lets you choose from among Relief, Emboss Over, Sobel, and Laplacian types of embossing.
- **Power:** Lets you adjust how pronounced the emboss effect is.
- **Angle:** Lets you choose the apparent angle at which the emboss effect is rendered.

Channels

Three checkboxes let you choose to use the Red, Green, and Blue channels to create this effect.

Mirrors

An effect that lets you reflect part of the image in any direction you like. At its simplest, this can create a split mirror effect like that found at the end of Prince's "When Doves Cry." At its most complicated, multiple reflections can be set up, to create whirling Kaleidoscope patterns in the image.

Main Controls

These controls let you choose what kind of effect this plugin has.

- **Mirror Placement:** Lets you choose different types of mirror effects including Individual (a simple reflection), Rosette, and Kaleidoscope.
- **Reflect at Borders:** Lets you add another reflection of the image whenever the angle and position of your reflection would result in offscreen black coming into the frame.

Individual Controls

When Mirror Placement is set to Individual, a series of six sets of Mirror parameters lets you add multiple mirrored edges in different combinations; by default only one is on, but you can turn more edges on to create more complicated mirror effects. Each group of Mirror parameters has the following parameters:

- **Enable Mirror:** Turns that mirror reflection on.
- **X and Y Position:** Positions the center of the mirror split reflection.
- **Angle:** Lets you adjust the angle of the seam along which a reflection will run.
- **Flip:** Flips which side of the image is reflected.

Rosette Controls

When Mirror Placement is set to Rosette, a single set of parameters creates a variety of circular patterns.

- **X and Y Position:** Positions the center of the circular rosette reflection.
- **Angle:** Lets you adjust the orientation of the rosette pattern about its center.
- **Wedge Width:** Lets you adjust the thickness of the wedges around the outside of the rosette pattern, and by extension how many wedges there will be. Thicker wedges results in fewer reflections around the rosette pattern and thinner wedges result in more reflections around the rosette pattern.

Kaleidoscope Controls

When Mirror Placement is set to Kaleidoscope, a single set of parameters lets you create a variety of kaleidoscopic images:

- **X and Y Position:** Positions the center of the mirror split reflection.
- **Center Size:** Lets you shrink or expand the cells comprising the kaleidoscopic image.
- **Angle:** Lets you adjust the angle of the seam along which a reflection will run.
- **Number of Sides:** Three sides produce a triangular pattern, while four sides produce a rectilinear pattern aligned with the currently selected angle.

Pencil Sketch in DaVinci Resolve (Studio Version Only)

A plugin with highly customizable controls for making an image look like it was drawn.

- **Color Sketch:** Check this box to toggle between a color or grayscale sketch.
- **Sketch Stroke Controls:** Parameters to control the thickness of outlines, the threshold at which they appear around objects in the frame, and their length.
- **Sketch Tone Controls:** Parameters to control how many levels of tonality are in the rendered result, and contrast controls to influence how much of the image falls into shadows, mid-tones, and highlights.
- **Sketch Texture Controls:** Parameters to apply a customizable stippled texture over the image, with a checkbox to auto animate it.



(Top) The original image, (Bottom) The image with Pencil Sketch applied

Prism Blur

Simulates flat chromatic aberration and a vignetted lens.

- **X and Y Position:** Let you change the center of the effect within the frame about which chromatic aberration is created.
- **Blur Strength:** Lets you blur the primary image.
- **Aberration Distance:** Adjusts how far the aberration effect appears from the source image.
- **Aberration Strength:** Intensifies the zoom blur used to create the chromatic aberration effect to intensify it.
- **Vignette Size:** Lets you add edge vignetting to the image; setting this to 0 removes the vignette, while 1 is the maximum size of a vignette using this filter.
- **Vignette Sharpness:** Lets you adjust the feathering of the vignette effect; lowering this value softens the edge, while raising this value sharpens, but never fully eliminates the softness from this edge. The Vignette effect is designed to fall off in a way that simulates how a lens would exhibit vignetting.

Scanlines

Simulates television scanlines, or any effect where you want alternating lines to darken the image. A variety of parameters makes this an extremely flexible effect that can be applied in numerous ways.

Appearance

Let you customize the type of scanline effect you want.

- **Line Frequency:** Lets you choose how many or few lines are superimposed on the image; fewer lines automatically space themselves to be thicker as they're uniformly distributed across the image.
- **Line Sharpness:** Lets you blur the border between each line, softening the effect.
- **Line Angle:** Lets you rotate the lines to appear at any angle; the default is 0 which results in horizontal lines.
- **Line Width:** Lets you increase or decrease the width of the blanking lines that obscure the image (which default to black but are tinted by the Color 2 parameter), thus increasing or decreasing the amount of image showing through.
- **Line Shift:** Lets you offset the lines being overlaid on the image. Animating Line Shift lets you create a rolling scanlines effect.

Color

Lets you tint and shift channels of the alternating scanlines created by this filter. Exaggerated, this can create numerous other stylistic effects.

- **Color 1 and Color 2:** Color picker and eyedropper controls that let you tint each alternating set of lines (eyedroppers let you sample a color from the RGB image being input into the current node).
- **Red, Green, and Blue Offset:** Lets you offset individual channels.
- **Scanlines Only:** Lets you see just the scanlines in isolation while you adjust them.

Composite

Lets you choose how to composite the scanlines effect against the original image.

- **Scanlines Only:** Lets you output an image consisting only of the scanlines patten you've created. Good for creating different kinds of patterned textures and mattes.
- **Composite Type:** Lets you choose which composite mode is used to blend the scanline effect into the image.

Sky Replacement

Replacing a drab, colorless sky with a more vibrant one or matching skies shot on different days are common colorist tasks that are made more efficient with the Sky Replacement effect. The benefits of using Sky Replacement instead of traditional compositing methods are a focused specific toolset allowing you to:

- Dynamically match the foreground to the brightness and color cast of the sky.
- Use Tracking options to move the sky along with the clip's camera movement.
- Add realistic computer generated sky elements, like sky color, horizon haze, clouds, and light sources.



(Top) The original bland sky, (Bottom) The image with a new better looking sky.

Selecting the Sky to be Replaced

It should be noted that the Sky Replacement effect does not do any sky selection itself, but rather takes that selection from other tools and focuses on blending the selection seamlessly with either a background plate of another sky or its own artificial sky generator. The benefit of separating the sky selection and the replacement tools is deliberate, as qualifying a sky is usually trivial, while getting a good mask at the horizon is not. This allows the colorist to use the right qualifier (HSL, Window, Magic Mask, etc.) for the particular scene, and then pass that on to the Sky Replacement effect. Another benefit of this arrangement is that the Sky Replacement effect can be applied to the same node as your qualifier, letting you select and replace the sky all in one node, and both tools can be updated concurrently as you use them.

To select the sky to be replaced and connect it to the Sky Replacement FX:

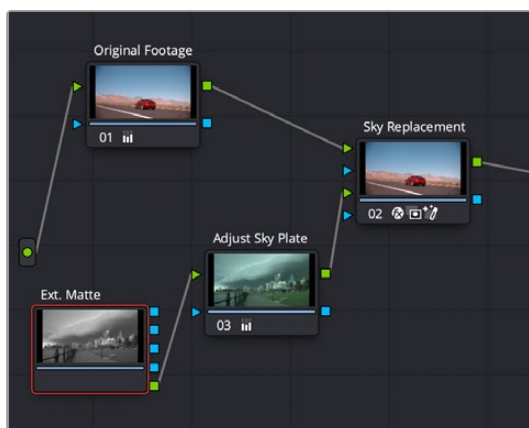
- 1 In a color node, use any Qualifier (Power Window, HSL, 3D, Magic Mask, etc.) to get a clean key of the sky and horizon.
- 2 Then drag the Sky Replacement effect from the Effects Library onto the same node.
- 3 Or create a new color node, drag the alpha output from the previous node (blue square), to the alpha input of the new node (blue triangle), and drag the Sky replacement effect from the Effects Library to the new node.

Selecting the Replacement Sky

Once your original sky is qualified, you have two choices on what to replace the sky with, either an artificially generated sky (created in the Sky Replacement Parameters below), or new footage of a previously shot sky as a background plate.

To connect a sky background plate to the Sky Replacement FX:

- 1 In a color node, qualify the sky and add the Sky Replacement FX.
- 2 Right-click on the node with the Sky Replacement FX, and select Add OFX Input from the drop-down menu. An additional color and alpha input appear on the node.
- 3 Drag the clip of the sky background plate from the Media Pool into the Node tree to make a new external matte of the clip.
- 4 Add a serial node off the external matte to make a new color node, so you can reposition and grade the sky background plate as necessary.
- 5 Connect the color output of the new color node (green square) to the Sky Replacement color input (the lower green triangle on the node). This will now use the video of the background plate for the sky replacement.



The node tree showing how to add your own sky background plate footage to the Sky Replacement effect.

Sky Mask Adjustments

These options let you make final adjustments to the source mask from whatever qualifier you've used.

- **Preview Mask:** Check this box to see the mask as a black and white matte; white is where the new sky will come through.
- **Shift Edge:** Shifts the mask boundary into, or away from, the horizon.
- **Refine:** Adjusts the mask boundary to the image edges.
- **Black Level:** Adjusts the transparency level of the mask.
- **White Level:** Adjusts the opacity level of the mask.

Source Sky Appearance

These tools let you customize the look of the new incoming sky. They have no effect on the artificial sky, but only on a video clip used as a sky background plate.

- **Color Space:** Lets you adjust the incoming sky's color space. Defaults to timeline settings.
- **Gamma:** Lets you adjust the incoming sky's gamma. Defaults to timeline settings.
- **Detail:** Lets you adjust the incoming sky's midtone detail.
- **Brightness:** Lets you adjust the incoming sky's brightness.
- **Saturation:** Lets you adjust the incoming sky's saturation.
- **Temperature:** Lets you adjust the incoming sky's color temperature.
- **Tint:** Lets you adjust the incoming sky's tint.
- **Opacity:** Lets you adjust the incoming sky's opacity, if you want to blend it in with the sky from the original footage.



The same car shot with a more dramatic storm footage used as the replacement sky, adjusted with the Source Sky Appearance tools.

Artificial Sky

These options let you generate realistic looking artificial sky elements. They are broken into three categories: Sky, Cloud, and Hotspot (stand in for the sun or moon). By default, the opacities for these elements are all set to zero (off). In order to see them, you must increase the opacity for each category to a positive number.

- **Preview Artificial Sky:** Checking this box shows all the Artificial Sky categories alone, without being composited with the foreground plate.

- **Sky Opacity:** Lets you adjust the opacity of the sky. Zero is totally off, and one is completely opaque. Any value in between is used to blend the artificial sky in with the source sky.
- **Sky Color:** Lets you set the color of the sky.
- **Horizon Color:** Lets you set the color of the horizon haze that blends with the sky color.
- **Horizon Softness:** Lets you adjust how soft the horizon blends into the sky.
- **Horizon Height:** Lets you set how high the horizon reaches in the frame.
- **Horizon Angle:** Lets you adjust the angle of the horizon.
- **Cloud Opacity:** Lets you adjust the opacity of the clouds in regards to the sky. Lower values blend the clouds in more; higher values make them stand out.
- **Cloud Color:** Lets you set the color of the clouds.
- **Cloud Scale:** Lets you set the size of the noise pattern that creates the clouds, letting you choose how close the clouds are to the foreground.
- **Cloud Shape:** Lets you adjust the horizontal/vertical ratio of the clouds. Lower values equal puffier (cumulus) clouds, while higher values make them more lengthened (cirrus).
- **Cloud Tilt:** Lets you adjust the angle of the sky relative to the ground, to match the foreground angle.
- **Cloud Detail:** Lets you determine the ratio of large to fine detail in the clouds.
- **Cloud Fill:** Lets you adjust the amount of cloud cover in the sky.
- **Cloud Contrast:** Lets you adjust how contrasty the cloud cover is compared to the sky.
- **Cloud Time:** Lets you change the noise pattern of the clouds. Using this control with keyframes will let you change the cloud pattern over time.
- **Hotspot Brightness:** Lets you adjust the intensity of the hotspot. Zero is no hotspot. There is an on-screen control for this option in the Viewer.
- **Hotspot Color:** Lets you adjust the color of the hotspot.
- **Hotspot Position:** Lets you adjust the position of the hotspot. There is an on-screen control for this option in the Viewer.
- **Hotspot Size:** Lets you adjust the size of the hotspot.
- **Hotspot Sharpness:** Lets you adjust the sharpness of the edges of the hotspot.

Sky Position

These tools let you track the motion in the foreground clip, in order to realistically move the sky along with any camera movement.

- **Match Motion:** Lets you choose which tracking method to use.
 - Keyframing Only:** Lets you manually adjust the sky movement by keyframing the position controls.
 - Track Foreground:** Automatically performs a track based on the foreground elements in the composite.
 - Track Original Sky:** Automatically performs a track based on the sky in the original clip. Only works when the sky has existing details like clouds. This is the best and most reliable tracking method to use if your footage allows.
 - Use FX tracker:** Links to the track generated by the FX tracker in the tracker controls.
- **Adjust Size:** Lets you adjust the size of the sky, so that it can cover the range of motion defined in the tracker.
- **Adjust Position:** Lets you manually adjust the X and Y positions of the sky.
- **Auto-Size for Motion:** Analyzes the track and automatically sets the size and position of the sky to encompass the track's extents.
- **Flip Horizontal:** Flips the sky on its horizontal axis.

Sky Integration

These controls help you match the attributes of the final sky to the foreground footage.

- **Preview Final Sky:** Check this box to view only the final sky output, including motion and any keyframed parameters.
- **Lens Distortion:** Adjust this to warp the sky to match the curvature of the original camera lens.
- **Lens Defocus:** Adjust this to move the sky out of focus and match the depth of field of the original lens.
- **Exposure:** Lets you adjust the overall brightness of the sky.
- **Add Grain:** These controls are a simplified set of tools to emulate the grain patterns found on motion picture film. Check this box to turn the grain pattern on or off.

Grain: Determines the amount of grain to add in the sky.

Size: Determines the size of the grain in the sky.

Softness: Determines how crisp the individual grains are.

Saturation: Adds color to the grain. 0 is monochromatic.

Foreground Appearance

Turning on these controls automatically attempts to adjust the overall look of the foreground, based on an analysis of parameters of the final sky.

- **Adjust Foreground:** Check this box to activate the Foreground Appearance tools below.
- **Auto Adaptation:** Selects the method by which the Foreground Appearance analysis is generated.

None: No automatic settings are applied. You can manually adjust the parameters below.

Globally Applied: The color cast is calculated from the visible sky and then applied globally to the foreground, with no variations in brightness. However, the color cast will update dynamically if the sky changes over time.

Match Horizon: The color cast is calculated from the visible sky but is weighted towards the horizon, its distance set with the Localization control. This will make brighter/darker areas of the sky apply bright/darker changes to the foreground. For example, if you have a dark sky but a moon on the horizon, the foreground will be brighter around the moon than elsewhere. This also updates dynamically over time.

- **Adaptation Amount:** Lets you select how much adaptive color adjustment to make.
- **Localization (Match Horizon only):** Lets you determine how far or near the color cast reaches from the horizon.
- **Brightness:** Lets you adjust the foreground's gain or brightness.
- **Saturation:** Lets you adjust the foreground's saturation.
- **Temperature:** Lets you adjust the foreground's color temperature.
- **Tint:** Lets you adjust the foreground's tint.

Sky Replacement Workflow

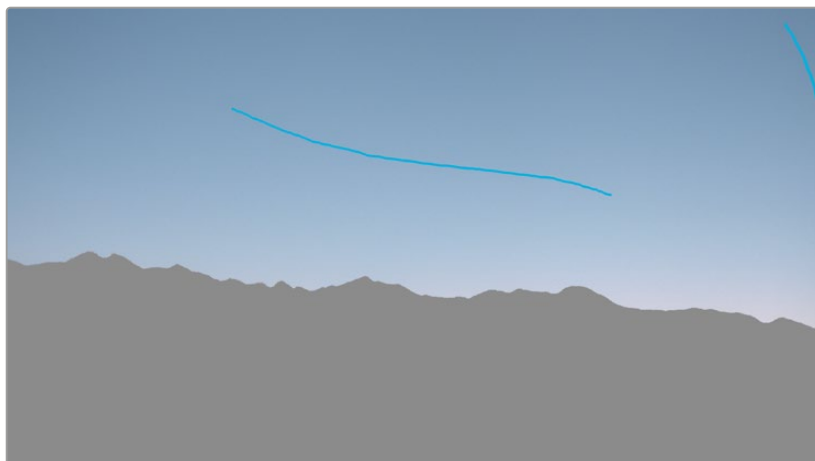
Here's a sample workflow for replacing the sky with one generated by the artificial sky tools.

This is the original clip, with the sky we want to replace. To start with, we simply make a new color node.



The original shot we want to replace the sky on.

Next we need to qualify the sky. For this clip, we can apply the Magic Mask (object) tool to the node to select just the sky we want to replace. You could also use the HSL qualifier, 3D qualifier, or even a power window to do the same thing.



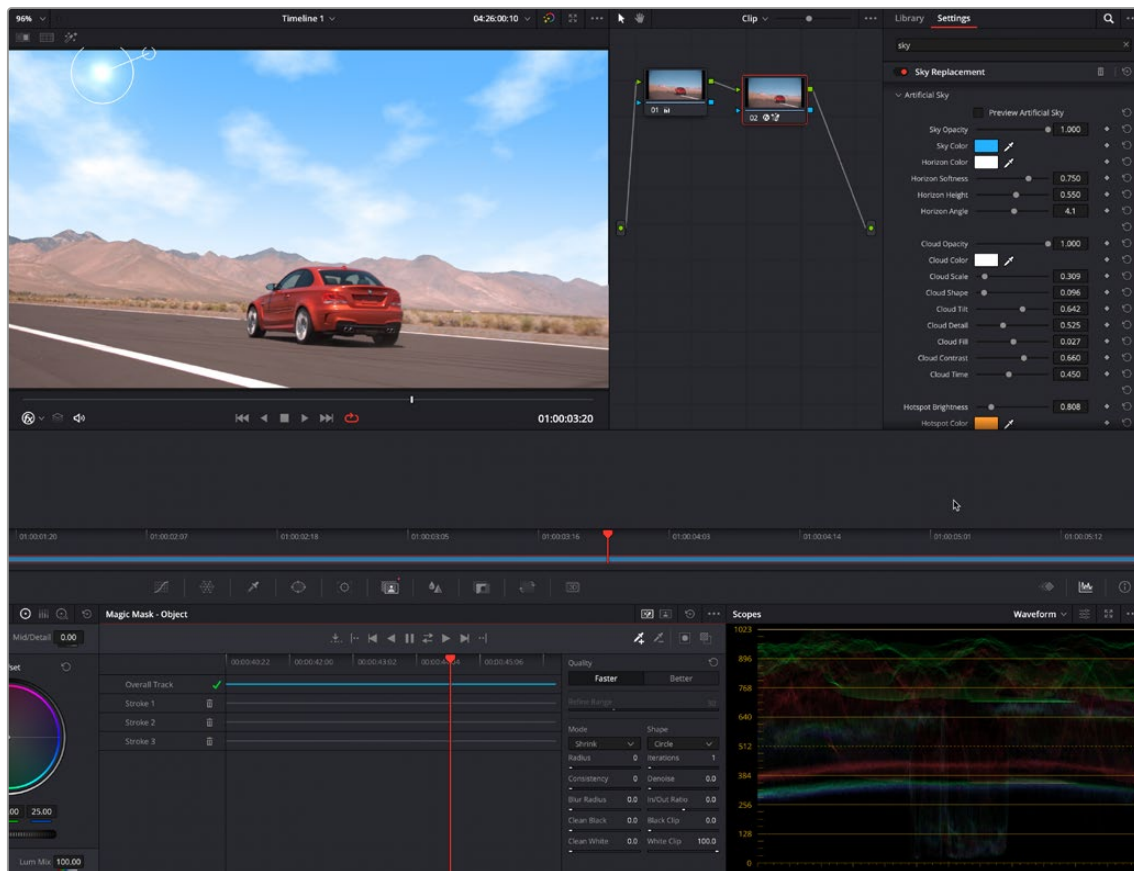
The sky qualified using the Magic Mask tool.

Now that the sky is selected, we can drag and drop the Sky Replacement effect from the Effects Library onto the same node that our Magic Mask is on.



The black area is where we will create our new sky.

Now that both toolsets are active on the same node, we can adjust both the Magic Mask and the Sky Replacement parameters at the same time. This makes the complete sky replacement a one node process.



You can use both the Magic Mask (or any qualifier) and the Sky Replacement tools on the same node.

Using the Artificial Sky Controls, we can create a new, more interesting sky. Then, using the Sky position tools, we can track the sky to the camera movement of the original clip. Finally, adding the Foreground Appearance toolset, we can seamlessly blend the foreground and sky together.



The final replaced sky that tracks with the camera movement.

Stylize (Studio Version Only)

A plugin that lets you apply one of a variety of painterly styles to an image based on analyses of different paintings, in such a way as to provide a temporally stable result for moving images as the style strokes will appear to be applied to individual objects in the scene that move and flow in a consistent way. There are two controls:

- **Styles:** A pop-up menu lets you choose an artistic style to apply to the image.
- **Style Scale:** This slider lets you adjust how large or small the applied art strokes should be applied.



(Top) The original image, (Bottom) The image with Style Transfer applied and set to Dance with a Style Scale of 4

TIP: This plugin is more flexible than you think, if you apply Style Transfer in a layered way against the original image. You can do this using a duplicate layer in the Edit page, or using two nodes representing the Style Transfer effect and original image connected with a Merge node in the Fusion page or a Layer node in the Color page. Once that's set up, simply combine the Style Transfer output with the original image using composite modes and opacity adjustments for more sophisticated blending. In particular, using the Luminosity composite mode lets you combine the texture from the Style Transfer while retaining the color of the original image, or using the Saturation composite mode lets you combine the color from the Style Transfer while retaining the texture of the original clip.

Tilt-Shift Blur (Studio Version Only)

Simulates depth-of-field effects using a progressive blur that's applied with a generated Z-depth map. The default settings create a “miniaturization” illusion, where the image appears to be tiny due to the top and bottom depth-of-field settings used.

Main Controls

These controls select and control the overall effect.

- **Blur Type:** You can choose between Fast Blur and Lens Blur (Lens Blur is the default).
- **Blur Strength:** A slider lets you adjust the amount of blur you add.

Lens Iris

The Lens Iris controls are only available when Blur Type is set to Lens Blur. This is a simplified set of controls from those of the Lens Blur filter.

- **Iris Shape:** Lets you choose what type of apertures you want to use, with which to influence the shape of the bokeh effect. Aperture Shape options include Triangle, Square, Pentagon, Hexagon, Heptagon, or Octagon shaped aperture.
- **Blade Curvature:** (Only available with Real and Creative Apertures) Lets you round off the edges of the Aperture Shape you selected.
- **Rotation:** Lets you adjust the angle the shape appears at.
- **Anamorphism:** Lets you adjust the aspect ratio of this effect in order to match the lens blur created by anamorphic lenses.
- **Highlights:** Lets you adjust how the highlights of the image affect the blur, dilating or eroding the image more or less depending on how high Smooth Strength is.

Depth of Field

Additional Depth of Field settings let you adjust the depth map being used to create the depth-of-field effect.

- **Map Source:** Lets you choose the source for the depth map. Custom (default) lets you create the depth map in the effect. From Alpha In and From Second Input allow you to import depth maps created in other nodes.
- **Center X, Center Y, and Angle:** These parameters let you transform the depth map.
- **Focus Sweep:** Lets you dial in a region based on “distance” for isolation. White areas of the depth map are affected by the blur, while black areas of the depth map are ignored.
- **In Focus Range:** Lets you expand or contract the black center of the depth map that defines the blurred region.
- **Near Blur Range and Far Blur Range:** Two parameters let you individually adjust the falloff of the white bottom and top of the depth map.
- **Depth Map Preview:** This checkbox lets you see the grayscale depth map directly while making these adjustments. The white area will be blurred, while the black area will be unaffected.

TIP: To create more believable Depth of Field adjustments, use the Depth Map effect in conjunction with the Tilt-Shift Blur effect. You can feed the results of the depth map directly into the Tilt-Shift Blur effect as a blur map, by connecting the key output of the Depth Map node to the key input of the Tilt-Shift Blur node, and selecting “From Alpha In” in the Map Source dropdown in the Depth of Field section of the effect. From there adjust the Focus Sweep to dial in the area you want to isolate. If you’re not getting the expected results, try inverting the depth map.

Watercolor (Studio Version Only)

A variation on the Abstraction filter, this filter reduces images into simplified washes of softly blended colors, in a very painterly fashion.

- **Channels:** This pop-up menu lets you choose whether to use Luminance or RGB to derive the smoothed color from.
- **Show Gradient:** This checkbox lets you see the boundaries that define each area of smooth color, based on the value you choose from the Smoothness slider below.
- **Smoothness:** Adjusts the amount of detail in the final result. Low smoothness values result in more detail in the smoothed result, while higher values result in larger pools of lower detail yet smoother color.